

week 2.

Buy side (order) → buy/sell

wants to generate profit
through trades.



decide what to buy, sell
and how much.

e.g. retail investor

• institutions.

sell side (quote) → price being offered

provides services to earn
a spread.



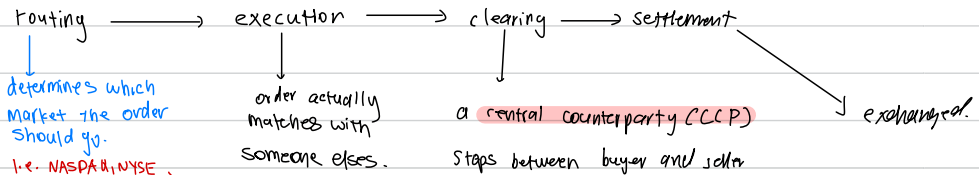
execute trades, provide research,
provide capital

e.g. brokers, dealers, market
makers.

bid: 99.5

ask: 100.5

life-cycle of an order:



Part 2 Auction vs dealerships

Auction (centralised + order driven) ASX NYSE

→ Buy and sell orders meet in a centralised system.

dealer (uncentralised + quote driven) NASDAQ, LSE

→ dealers quote prices (bid, ask) directly to the customer.

More on Auction (centralised) Markets.

call (batch) auction.

→ orders (buy/sell)
accumulate first and
then execute together

for example before market
opens all order accumulate
and then find at what
price can we execute
it to get the largest
amount of trading.

p^* Q^*

→ slowly just find
the optimal price to
have most volume,

→ and they all
execute at the
same price.

→ occurs at
market open, close
and halt

→ give most
sensible price.

continuous auction.

→ orders continuously arrive and
execute throughout the day

jump in after an
auction market

→ use continuous
auction and limit order
book.

Dealership side. you trade w/ a dealer

different dealers can quote different prices.

→ inter dealer market — you
lowkey can use other dealers
quotes to reduce the risk of
your inventory.

inventory risks.

① revaluation risk → price change

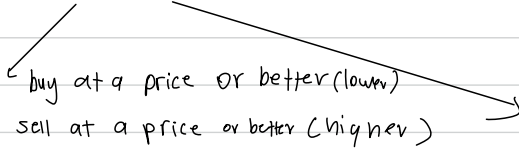
② carry costs → opportunity cost

quote skewing → change quotes depending
on risk.

Summary: Earn the spread by giving customers liquidity, but
every trade changes inventory so skew quote to change risk.

types of orders.

Market, limit, stop, iceberg.



iceberg/ DarkIce: where

You hide the volume of your order -

eg. you sell 2k shares but only show you sold 200 protects information -

Sell stop: share price = \$100
Sell stop = \$98

if price hits 98, it will become a market sell order -

difference in price is called
slippage -

buy stop: share = \$100

buy stop = \$102
if price reaches 102, buy
market order

HYBRID/ DUAL markets-

HYBRID (NYSE): central limit order book (auction) with market maker/ dealers.

LOB + market maker

DUAL (NASDAQ, LSE): same thing

4 dimensions of market design.

centralisation: auction vs dealer.

intermediation:

transparency:

trading frequency:

direct vs intermediated.

buyer $\rightleftharpoons$ seller

buyer $\rightleftharpoons$ dealer $\rightleftharpoons$ seller

pre-trade transparency

· bid/ask · volume · depth of order book.

post-trade transparency.

→ how fast it is reported.